PROJECT PROPOSAL

Smart Attendance and Examination Management System

QR Code-Based Web Application

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A practical web-based solution for QR attendance, examination scheduling, automatic attendance calculations, and academic reporting.

1. Introduction

The Smart Attendance and Examination Management System is a web-based application designed to manage student attendance and examination schedules. It will help teachers maintain student records and mark attendance using QR codes. Students will also be able to check their attendance and view upcoming examination schedules through the website.

2. Problem Statement

In many educational institutions, attendance and examination records are managed manually. This process takes time and may cause errors in record keeping and attendance calculations. Teachers may also face difficulties managing examination schedules and preparing attendance reports.

This project aims to solve these problems by providing a simple web-based system that makes attendance and examination management easier and more organized.

3. Objectives

- To develop a web-based attendance management system.
- To mark student attendance using QR codes.
- To manage student, teacher, and class records.
- To calculate attendance percentages automatically.
- To manage examination schedules.
- To generate attendance reports.
- To reduce manual work and improve record accuracy.

4. Proposed System

The proposed system will provide three main panels: Admin, Teacher, and Student. The admin will manage user accounts and class records. Teachers will be able to generate QR codes for attendance, view attendance records, and manage examination schedules. Students will be able to log in to the website to check their attendance percentage, attendance history, and upcoming examinations.

The system will store all records in a database and allow authorized users to access information according to their roles.

5. Main Features

Feature	Description
User Login	Separate access for admin, teachers, and students.
Student and Class Management	Maintain student details and organize students by class.
QR Code Attendance	Teachers create attendance sessions and students scan the session QR code.
Automatic Calculations	Calculate attendance percentages and display attendance history.
Examination Timetable	Add and view examination dates, subjects, and schedules.
Low-Attendance Alerts	Highlight students whose attendance falls below a set threshold.
Reports	View and download attendance reports for selected classes or date ranges.

6. Tools and Technologies

Component	Proposed Technology
Frontend	HTML, CSS, JavaScript
Backend	Python and Django
Database	PostgreSQL
QR Code	QR code generation and scanning library
Development Tool	Visual Studio Code
Version Control	Git and GitHub

7. Project Scope

This project will focus on attendance management and examination scheduling in educational institutions. It will provide a simple platform for teachers to manage attendance and exam records and for students to view their academic information.

The project will use QR codes for attendance instead of face recognition to keep the system manageable for an individual student project.

8. Expected Outcome

The expected outcome is a user-friendly web application that simplifies attendance management and examination scheduling. It will reduce manual work, improve the accuracy of attendance records, and make academic information easier to access.

9. Conclusion

The Smart Attendance and Examination Management System will provide a convenient solution for managing student attendance and examination schedules. By using QR codes, automatic calculations, and reporting in one web-based platform, the system aims to save time and improve record accuracy.